FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Obstetric Fistula (VVF/RVF)

Tutorial | Week 5 · Bleeding, cancer & fertility | Gynaecology | Dr Kavuma

An obstetric fistula is an abnormal opening between the genital tract and the bladder (VVF) or rectum (RVF). It is usually caused by ischaemic pressure necrosis from prolonged obstructed labour, and increasingly by surgical injury at CS or hysterectomy. It presents as continuous leakage of urine or faeces per vaginam. It is preventable by timely delivery and immediate catheterisation, and curable in most women by skilled, tension-free vaginal repair followed by uninterrupted bladder drainage.

Must know before the ward round

1. Commonest cause in Uganda is prolonged obstructed labour, where the impacted head crushes the bladder base and vagina against the pubis. The necrotic slough separates at about 3–10 days postpartum, so leakage starts days after delivery. Iatrogenic (CS/hysterectomy) fistulae now make up nearly a third of repairs.
2. Immediate leakage after surgery suggests a direct surgical injury. Leakage that starts days after delivery suggests an ischaemic (obstructed labour) fistula.
3. Continuous leakage with normal voiding suggests a ureterovaginal fistula. Cyclical haematuria with amenorrhoea and continence suggests a vesicouterine fistula (Youssef syndrome).
4. Dye test: instil dilute methylene blue into the bladder with swabs in the vagina. Blue = VVF. Wet but not blue = ureterovaginal (confirm with the three-swab or double-dye test).
5. A fresh fistula needs a Foley catheter on continuous free drainage for up to 4–6 weeks with high fluid intake. Small fresh fistulae may close without surgery.
6. Repair principles are good exposure, wide mobilisation, tension-free watertight closure in separate layers and uninterrupted bladder drainage afterwards (WHO 2018: 7–10 days for a simple fistula).
7. Goh classification uses the distance of the distal edge from the external urinary meatus (type 1 more than 3.5 cm to type 4 less than 1.5 cm), size (a–c) and scarring (i–iii). Type 4 (urethral involvement) has the worst continence outcome.
8. After repair: no intercourse for about 3 months, family planning, and deliver every future baby by elective CS in hospital.
9. Rule out the mimics first: palpate the bladder and check a post-void residual (over 150 mL = retention with overflow), and do a dye test before labelling any leakage a fistula.
10. Three-swab test: upper swab wet but not blue = ureterovaginal; middle swab blue = mid-vaginal VVF; upper swab blue = juxtacervical or vault; lowest swab only = urethral (or dye spill).
11. Always document the distance from the external urethral meatus, the size, the scarring, the urethral length and a rectal examination, so the fistula can be Goh-graded and triaged as simple or complex.

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1. Definition

- **Obstetric fistula** is a hole between the vagina and the **bladder or urethra** (vesicovaginal fistula, **VVF**) or the **rectum** (rectovaginal fistula, **RVF**). It is usually caused by **days of obstructed labour** and, increasingly, by **surgical injury at caesarean section or hysterectomy**.
- The result is **continuous, uncontrollable leakage of urine and/or faeces**. That leads to wet, smelly clothes, **ammoniacal dermatitis**, and then **divorce, stigma, isolation, poverty and depression**. The baby has usually died in the labour that caused the fistula.
- **Each fistula is a failure of the health system**: a delay in deciding to seek care, in reaching care, or in receiving a CS.
- The **one thing never to forget**: after any prolonged or obstructed labour, **put in a catheter and leave it for 7–14 days**. For a fresh fistula, **continuous bladder drainage for up to 4–6 weeks** can close a small one without surgery.
- Most fistulae can be **repaired vaginally** by a trained surgeon, with closure rates of about 80–90%. Residual **stress incontinence** remains a common problem, especially when the urethra is involved.

Types by anatomy

Type	Communication	Typical clue
Urethrovaginal	Urethra to vagina	Leakage may be positional; the closing mechanism is involved, so there is a high risk of post-repair stress incontinence
Vesicovaginal (VVF)	Bladder to vagina (mid-vaginal, juxtacervical, vault)	**Continuous leakage day and night**; little or no voiding
Vesicocervical / vesicouterine	Bladder to cervix or uterus, usually after CS	**Youssef syndrome**: cyclical haematuria (menouria), amenorrhoea, and often continence
Ureterovaginal	Ureter to vagina, usually after CS or hysterectomy	Continuous leakage **plus normal voiding** (the other ureter still fills the bladder). May have loin pain or hydronephrosis
Rectovaginal (RVF)	Rectum to vagina	Flatus or faeces per vaginam. It may follow obstructed labour, **third- or fourth-degree tears** (low RVF), or sexual violence
Combined VVF + RVF	Both	Severe obstructed-labour injury complex

2. Epidemiology

- WHO estimates **50,000–100,000 new obstetric fistulae worldwide each year**. Recent estimates suggest about half a million women are living with fistula, down from about 2 million two decades ago, but still overwhelmingly in **sub-Saharan Africa and South Asia**.
- **Uganda**:
  - The **Uganda Demographic and Health Survey (UDHS)** shows the proportion of women reporting fistula symptoms falling from about 3% (2006) to about 1% (2016). The highest burden is in the **Tooro and Ankole** regions.
  - Ministry of Health estimates have ranged from **about 100,000 to 200,000 women living with fistula**, with **about 1,900 new cases a year**.
  - A multicentre Ugandan cohort of **510 repairs (2010–2021)** found **66% ischaemic** fistulae and nearly **one third iatrogenic or other**. Successful **closure with continence was about 81%**.

- **The rising iatrogenic share** reflects more CS being done, often late, on tired surgeons and oedematous tissue, by less experienced hands at district level. Every CS for obstructed labour is also a risk to the bladder.

> **UGANDA CONTEXT**
>
> Uganda treats fistula through **regional fistula centres and outreach "fistula camps"**. About **29 centres across 24 districts** have been set up, including national and regional referral hospitals and mission hospitals.
>
> - **Kitovu Hospital (Masaka)** runs camps several times a year.
> - **TERREWODE** (Soroti) runs a dedicated **women's community fistula hospital** and a reintegration programme.
> - The **Fistula Foundation Treatment Network** was launched in Uganda with the Ministry of Health and partners.
>
> Surgery is usually **free** to the woman, funded by the MoH, UNFPA and NGOs. Your job on the ward is to **identify, catheterise, counsel and link her** to the nearest centre. Know when the next camp is and who the regional fistula surgeon is.

3. Aetiology and Risk Factors

Causes

Cause	Mechanism	When leakage starts
Ischaemic (obstetric)	Prolonged obstructed labour, where the head compresses soft tissue against the bone, causing **pressure necrosis**	**3–10 days postpartum**, when the slough separates
Iatrogenic (surgical)	Bladder injury at **CS** (especially repeat, emergency or for obstructed labour), **caesarean hysterectomy**, abdominal or vaginal hysterectomy, **destructive operations**, forceps, symphysiotomy	Often **immediate or within 1–2 days** of surgery; ureteric injuries after 1–2 weeks
Traumatic	Sexual violence, penetrating injury, harmful traditional practices (e.g. gishiri cuts), foreign bodies	Variable
Other	Cervical cancer, **radiation**, schistosomiasis, LGV, TB (rare)	Gradual

Risk factors (why women still get fistula): child marriage and adolescent pregnancy, short stature and a small pelvis, home delivery, poverty, distance and transport problems, poor labour monitoring, delayed referral, and HC IIIs without theatres. For iatrogenic fistula: late CS for obstructed labour, repeat or emergency CS, oedematous tissue, tired or inexperienced surgeons, operating without an empty bladder or proper bladder flap.

4. Pathophysiology

The obstructed labour injury

1. The fetal head is **impacted in the pelvis for hours to days**.
2. Soft tissues trapped between the head and the **pubic symphysis** (anteriorly) or the **sacrum** (posteriorly) lose their blood supply. This is **ischaemia**.
3. The **bladder base, trigone, bladder neck, urethra** and anterior vaginal wall die (pressure necrosis). Posteriorly, the **rectum** can necrose.
4. Over **3–10 days** the dead tissue sloughs, leaving a **hole surrounded by oedema and infection**. The edges then **scar (fibrose)** over weeks.

5. Where the head was impacted predicts where the fistula forms:
    - At the **inlet**: high, juxtacervical or vesicouterine injury.
    - At the **mid-pelvis**: mid-vaginal VVF, the classic type.
    - At the **outlet**: urethral or bladder-neck injury, plus RVF.

The "obstructed labour injury complex" (Arrowsmith): a field injury, not just a hole. The same ischaemia damages much more than the bladder:

- **Urethral loss** (causes stress incontinence even after closure).
- **Vaginal scarring and stenosis** (painful sex, infertility).
- **Ureteric involvement**.
- **Foot drop** (lumbosacral trunk or common peroneal nerve injury).
- **Pubic bone osteitis**.
- **Secondary amenorrhoea** (stress, malnutrition, Sheehan's syndrome after PPH).
- **Chronic dermatitis**.
- **Bladder stones**.
- **Infertility**.
- **Stigma, depression and poverty.**

This explains why "the fistula is closed but she still leaks": the continence mechanism (urethra and bladder neck) was destroyed along with the bladder base.

Why catheter drainage can heal a fresh fistula. A **decompressed bladder** does not stretch the defect, and urine is diverted away from the wound. The raw, granulating edges can then **contract and epithelialise**. This works only while the fistula is **small and fresh**. Once the edges are fibrotic and epithelialised (after weeks), spontaneous closure is unlikely.

Why surgery follows the principles it does

- **Tension-free closure**: tension causes ischaemia, which breaks the suture line down.
- **Wide mobilisation** of the bladder from the vagina allows tension-free closure.
- **Separate, non-overlapping layers** (bladder, then vagina): reduce re-fistulation.
- **Well-vascularised tissue** (e.g. a **Martius fat-pad flap** in scarred cases): brings in blood supply.
- **Uninterrupted drainage**: a **blocked catheter** lets the bladder distend and blows the repair apart. That is why nurses check the catheter hourly.

Prolonged obstructed labour: head impacted for hours to days

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Soft tissue crushed against pubis or sacrum

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Ischaemia of bladder base, urethra, vagina, rectum

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Pressure necrosis

Field injury: urethral loss, vaginal stenosis, foot drop, amenorrhoea

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Slough separates at 3-10 days

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Hole: VVF, urethral fistula, RVF

Edges fibrose over weeks: spontaneous closure unlikely

↓

Continuous leakage of urine or faeces

↓

Dermatitis, UTI, stones, stigma, depression

Feature links

- Necrosis takes days to slough → leakage starts **3–10 days** after delivery (ischaemic).
- Direct surgical injury → leakage **immediately** or within 1–2 days of surgery.
- Ureteric injury with one healthy ureter → continuous leakage **plus normal voiding**; ureteric obstruction → loin pain, hydronephrosis, fever.
- Bladder–uterus communication with a continent bladder neck → menstrual blood into the bladder → **cyclical haematuria (menouria), amenorrhoea, continence** (Youssef).
- Urine on the skin → ammonia → **ammoniacal dermatitis**.
- Urethral/bladder-neck destruction → **stress incontinence** after closure.
- Lumbosacral trunk or common peroneal compression by the head → **foot drop**.
- Vaginal ischaemia and scarring → **stenosis**, dyspareunia, infertility.
- Women drink less to leak less → concentrated urine → **UTI, stones**, malnutrition.
- Rectal necrosis or missed third/fourth-degree tear → flatus/faeces per vaginam (**RVF**).
- Bladder injury leaking into the peritoneum → **urinary ascites**, peritonism, sepsis.

5. Classification / Staging

Waaldijk classification (surgical)

- **Type I**: does **not** involve the urethral closing mechanism.
- **Type II**: **involves the closing mechanism**.
  - **A**: without (sub)total urethral involvement. **B**: with (sub)total urethral involvement.
  - **a**: without a circumferential defect. **b**: with a circumferential defect.
- **Type III**: ureteric and other exceptional fistulae.

Surgical difficulty rises, and closure and continence rates fall, from **type I through type IIBb**.

Goh classification (predicts outcome better)

Component	Grade	Definition
Site (distal edge from the external urinary meatus, EUM)	Type 1	More than **3.5 cm**
	Type 2	2.5–3.5 cm
	Type 3	1.5 to under 2.5 cm
	Type 4	**Under 1.5 cm** (involves the urethra)
Size (largest diameter)	a	Under 1.5 cm
	b	1.5–3 cm
	c	Over 3 cm
Scarring	i	None or mild fibrosis; vaginal length over 6 cm, normal capacity
	ii	Moderate or severe fibrosis; reduced vaginal length or capacity
	iii	Special considerations: post-radiation, ureteric involvement, circumferential defect, previous repair

For example, a "**Goh 2bi**" fistula has its distal edge 3 cm from the EUM, is 2 cm across, and has minimal scarring. It has a good prognosis. A **type 4** fistula (with the urethra involved) is the most likely to fail or leave stress incontinence.

EXAM FAVOURITE

Simple vs complex fistula (WHO/FIGO working definition). A **simple** fistula is single, small (under about 3–4 cm), not involving the urethral closing mechanism, not circumferential, with minimal scarring, no ureteric involvement, and a first repair. It can be repaired by a trained generalist. A **complex** fistula (large, urethral, circumferential, scarred, previous failed repair, combined VVF/RVF or ureteric) needs a **fistula specialist centre**.

6. Clinical Features

History

Fistula is usually a **planned, dignified outpatient or ward workup**, not a resuscitation. The history answers: is it really a fistula, which organ is leaking, how fresh is it, and is she supported for repair?

Ask	Why it changes the plan
The index labour: how long, where, how delivered (SVD, CS, destructive), baby alive?	Days of labour + stillbirth + CS or destructive delivery = **ischaemic** fistula; CS or hysterectomy = consider **iatrogenic** (bladder or ureter)
When did the leaking start?	**Immediately** after surgery = direct surgical injury (bladder or ureter); **3–10 days** after delivery = ischaemic slough; **1–3 weeks** after CS/hysterectomy = think **ureter**
Continuous, or only on coughing or standing, or with urgency?	Continuous = fistula; effort only = stress incontinence; urgency = overactive bladder or UTI
Does she still void normally?	**Continuous leakage + normal voiding = ureterovaginal** fistula (the other ureter fills the bladder)

Ask	Why it changes the plan
Faeces or flatus per vaginam? Ask directly, because women may be ashamed	**RVF**, or combined VVF/RVF; changes the operation (bowel prep, possible colostomy)
Periods; blood in the urine with periods	Cyclical haematuria + amenorrhoea + continence = **vesicouterine** (Youssef); amenorrhoea is also common after obstructed labour (stress, malnutrition or Sheehan's)
Weak leg, dragging the foot	**Foot drop**: part of the injury complex; physiotherapy before surgery
Previous repair: when, where, how many	Scarring, lower success with each attempt: **specialist centre**
Is she pregnant or breastfeeding? Contraceptive and pregnancy wishes	Pregnancy test before surgery; plan contraception so she does not conceive before healing; timing of repair and future delivery planning
Fluid intake	Many women drink less to leak less: dehydration, concentrated urine, UTI, stones
Social: husband, family, income, livelihood, isolation, where she sleeps, transport	Counselling, reintegration and transport support; who will stay with her during the catheter period
Mood, suicidal thoughts	Depression is very common; suicidal thoughts are a red flag needing same-day support
HIV status, TB, chronic illness, nutrition, anaemia	Affect healing and fitness for anaesthesia

Examination

Follow the order of Hutchison's Clinical Methods (24th edition). If she is febrile, peritonitic or anuric after a CS or hysterectomy, this is a surgical emergency (section 9).

Introduction, consent and preparation

- **Greet her privately and with dignity.** Offer a pad and a seat she will not worry about wetting. Many women have been sent away or mocked; the first minute decides whether she tells you the whole story.
- Introduce yourself, explain, gain consent; a **chaperone**.
- **Hand hygiene**.
- **Position and exposure**: supine for the abdomen; **left lateral (Sims) or lithotomy** for the vaginal examination, with good light (a **head torch**).

General inspection and vital signs

- Nutrition (weight, MUAC), **pallor**, hydration, mood, and the **smell of urine**.
- **Gait**: look for **foot drop** (a high-stepping gait, weak ankle dorsiflexion).
- **Vitals**: temperature, pulse, BP, RR. **Fever, tachycardia or hypotension** in the first weeks after obstructed labour or CS: look for **puerperal sepsis, pelvic abscess, necrotic slough** or **urinary ascites**.

Hands, face and neck

- Conjunctival and palmar pallor; dry mucous membranes (women drink less to leak less).

Abdomen

- **Inspection**: CS scar, distension.
- **Palpation**: **palpable bladder** (retention with overflow, not a fistula), suprapubic tenderness, peritonism (urine or faeces leaking into the peritoneum), **renal angle tenderness** or a flank/**loin mass** (obstructed ureter or urinoma after CS or hysterectomy).

Vulva and perineum

- **Ammoniacal dermatitis** (red, excoriated, sometimes with papules or hyperkeratosis) of the vulva and thighs, wetness, pooling urine, perineal body present or absent, any scar from a tear or episiotomy, a gaping anus.

Speculum and digital examination

Use a **Sims speculum** in the left lateral position (or lithotomy). Document:

Document	How	Why
Site	Juxtaurethral, mid-vaginal, juxtacervical, vault, cervical	Predicts route and difficulty
Distance of the distal edge from the external urethral meatus (EUM)	Measure in cm with a marked sound or finger	Goh type 1 (>3.5 cm) to type 4 (<1.5 cm)
Size	Largest diameter in cm (or fingertips)	Goh a (<1.5 cm), b (1.5–3 cm), c (>3 cm)
Scarring and vaginal length/capacity	Rigid, fixed edges; vagina >6 cm or shortened	Goh i–iii; need for a Martius flap
Urethra	Pass a **metal catheter or sound** up the urethra: is it intact and how long? Does the tip appear through the fistula?	Urethral involvement predicts **residual stress incontinence**
Edges and mobility	Slough, granulation, infection, or clean epithelialised edges; feel the rim and how mobile the tissues are	Fresh + small = catheter trial; clean = ready for repair
Cervix	Present, torn, ulcerated or a mass	A friable hard mass = **biopsy** (cancer)
Ureteric orifices / bladder mucosa	Seen through a large fistula; prolapsing bladder mucosa	Near the trigone = ureteric catheters at repair

Rectal examination (never skip)

- An **RVF** (site, size, distance from the anal verge), **anal sphincter** tone and continuity (look for a cloaca, or a missing perineal body).

Neurological examination of the legs

- Ankle dorsiflexion power, sensation over the dorsum of the foot (common peroneal) and lateral leg.

Completing the examination

- Bedside **dye test** (section 8), thank her, hand hygiene, summarise with a Goh grade and a simple/complex triage.

Always document the distance from the EUM, the size, the scarring, the urethral length and the rectal findings, so the fistula can be Goh-graded and triaged as simple or complex.

Red flags

RED FLAG

- **Fever, loin pain or a flank mass after hysterectomy or CS**: ureteric injury with obstruction or urinoma.
- **Sepsis or peritonitis** after a difficult CS: bladder or bowel injury with urine or faeces leaking into the peritoneum.
- **Anuria**: bilateral ureteric injury or AKI. This is an urgent urology problem.
- **Suicidal ideation**, or severe depression.
- **Necrotic, foul slough** with systemic illness: needs debridement and antibiotics.
- **A fistula with an ulcerated, friable, hard mass**: **cervical cancer** until proven otherwise. Biopsy it.

RED FLAG

Fever, peritonism, loin pain or anuria after a CS or hysterectomy is a **surgical emergency** (bladder, bowel or ureteric injury). This is not a "fistula for the next camp": call the senior surgeon and urologist.

Thinking like a clinician

1. **"She leaks since delivery." Is it really a fistula?** Separate **continuous** leakage (fistula), **stress** leakage (weak pelvic floor or urethral damage), **urgency** leakage (overactive bladder, UTI) and **overflow** (retention after delivery, epidural, pain). **Overflow incontinence** after a prolonged labour is common and treatable with a catheter. Always palpate for a **distended bladder** and do a **dye test** before labelling a fistula.
2. **"The dye test is negative but she leaks continuously"**: think **ureterovaginal fistula**. Do the three-swab or double-dye test and scan the kidneys.
3. **"She is continent but has blood in her urine with her periods"**: a **vesicouterine** fistula, usually after a CS.
4. **"The fistula is fresh"**: this is an **opportunity, not a disappointment**. Catheterise now. Don't send her home to wait 3 months without a catheter.
5. **Fistula is a symptom of the whole system**: document the delays (the three-delay model) and report the case for MPDSR-style review.
6. **Common student mistakes**:
    - forgetting the **rectal examination**
    - not asking about **faecal leakage**
    - not measuring the **distance from the EUM**
    - ignoring **foot drop** and **mood**
    - calling any post-delivery leakage a "VVF" without a dye test

CLINICAL PEARL

Always do a dye test before you call it a fistula. Postpartum **overflow incontinence** from a distended, atonic bladder looks exactly like a small VVF and is cured by a catheter for a few days.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
VVF	Continuous leakage; blue dye seen in the vagina	Dye test; speculum
Ureterovaginal fistula	Continuous leakage **plus normal voiding**; after hysterectomy or CS; hydronephrosis	Three-swab test (swabs wet but not blue), double-dye (orange), renal US, IVU/CT urogram

Differential	Distinguishing clinical feature	Distinguishing investigation
Vesicouterine fistula	Cyclical haematuria, amenorrhoea, often continent; after CS	Cystoscopy, hysterography, dye appearing from the os
Stress urinary incontinence	Leakage only with cough, sneeze or effort; dry at rest	Cough stress test with a full bladder; negative dye test
Overflow incontinence (postpartum retention)	Distended bladder, dribbling, poor stream	Palpation or bladder scan; residual over 150 mL on catheter
Overactive bladder / UTI	Urgency, frequency, dysuria	Urinalysis, culture
Vaginal discharge (vaginitis, lochia)	Not urinous, no ammonia smell	Speculum, swab, dye test negative
RVF vs anal incontinence	Faeces per vaginam (RVF) vs per anus without control (sphincter tear)	Speculum, rectal exam, methylene blue enema or tampon test
Malignant or radiation fistula	Hard, friable, ulcerated, bleeding edges; radiation history	**Biopsy**

8. Clinical Workup

The workup answers five questions: **Is it really a fistula (or overflow, stress or urge leakage)? Which organ is leaking (bladder, urethra, ureter, uterus, rectum)? Where is it and how big and scarred is it (Goh)? Is it fresh enough for catheter drainage? Is she fit and supported for repair?**

Bedside

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Test	How	What it means	Available at
Palpate / catheterise the bladder	Catheter after she voids; residual volume	**Residual over 150 mL** = retention; dribbling may be **overflow**, not a fistula	All levels
Dye test	**Dilute methylene blue (about 100–200 mL)** into the bladder through a Foley; watch with a speculum, or place a gauze in the vagina	**Blue fluid in the vagina = VVF**; note exactly where it appears; separates fistula from other leakage	HC III and above
Three-swab (tampon) test	Worked example 1 below	Tells **VVF vs ureterovaginal vs urethrovaginal**	HC III and above
Double-dye test (where phenazopyridine is available)	Oral phenazopyridine (renal urine turns orange) + methylene blue in the bladder	Orange swab = ureteric; blue = bladder; both = combined	Where available
Cough stress test (full bladder, no fistula leak seen)	Ask her to cough while you watch the meatus	Leakage from the meatus with cough = **stress incontinence**	All levels
Urine dipstick	Nitrites, leucocytes, blood	UTI (treat before surgery); blood = stones, schistosomiasis or vesicouterine fistula	All levels
Pregnancy test	Urine hCG	Before surgery	All levels

Test	How	What it means	Available at
HemoCue / Hb, RBS	Finger-prick	Anaemia and diabetes affect healing	HC III and above
HIV and syphilis RDT	If status unknown	Healing, counselling, ART	All levels
Rectal methylene blue or tampon test (suspected small RVF)	Tampon in the vagina, dilute blue (or air with the vagina under water) instilled per rectum	Blue on the tampon, or bubbles = RVF	HC IV and above

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
Urine microscopy and culture	Everyone before repair; symptoms of UTI	Pyuria, organism; **schistosome ova** in endemic areas	Treat per culture before surgery; **praziquantel 40 mg/kg** if ova	Microscopy at HC IV; culture at regional/national referral hospitals
FBC / Hb	Everyone before repair	Anaemia (common: blood loss, poor diet)	Iron and folate; transfuse if very low	HC IV and above
Stool for ova / blood	Combined RVF, anaemia	Hookworm, schistosomiasis	Deworming (**albendazole 400 mg**) before surgery	HC IV and above
Urea, creatinine, electrolytes	Suspected ureteric injury, anuria, before repair at most centres	Raised creatinine	Suggests obstruction or bilateral injury: urgent imaging and urology	Regional/national referral

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Renal and pelvic ultrasound	Suspected ureteric fistula, loin pain, recurrent UTI	**Hydronephrosis**, **urinoma**, **bladder stones**	Ureteric injury = urology (stent or reimplantation); stones removed at repair	Most HC IVs and hospitals
IVU or CT urogram	Suspected ureterovaginal fistula (wet-not-blue swab)	Contrast leak from a ureter; obstruction	Plans stenting or reimplantation	Regional/national referral (CT at national referral and some private centres)
Hysterography / cystography	Suspected vesicouterine fistula	Contrast from uterus to bladder or vice versa	Abdominal repair	Referral centres

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Cystoscopy / examination under anaesthesia	Complex, high or recurrent fistulae; vesicouterine; before or at repair	Relation to the **ureteric orifices** and trigone; stones; extent	Surgical route and ureteric catheters	Fistula centres and regional referral hospitals

Test	When	Expected / abnormal finding	How it changes management	Available at
Biopsy	Friable, ulcerated or hard mass; radiation history	Carcinoma	Changes everything: oncology, not fistula repair	General/referral hospitals; histology at referral labs

Putting it together

Model one-line summary: "A 22-year-old para 1+0 with no living child, 5 weeks after a 3-day labour ending in a CS and a fresh stillbirth, with **continuous urinary leakage since day 6**, **no normal voiding**, **right foot drop** and ammoniacal dermatitis; a **positive dye test** and a **2 cm mid-vaginal VVF 3 cm from the EUM with mild scarring (Goh 2bi)**: an ischaemic VVF after obstructed labour."

Problem list

1. Ischaemic VVF, Goh 2bi (still within the 4–6-week catheter window).
2. Ammoniacal dermatitis.
3. Right foot drop (common peroneal / lumbosacral injury).
4. Anaemia (Hb 9.2 g/dL).
5. Bereavement, low mood, rejection by her husband.

Working diagnosis and plan: ischaemic VVF → Foley on free drainage, fluids 4–5 L/day, sitz baths and barrier cream, iron and folate, physiotherapy, counselling; re-examine in 1–2 weeks; refer to the fistula centre if not closed by 4–6 weeks.

Model case presentation (ward round)

> "Mrs A is a 22-year-old para 1+0 from Kasese, with no living children, who presents 5 weeks after delivering a fresh stillbirth by emergency CS at a general hospital after 3 days of labour. Since day 6 postpartum she has had **continuous leakage of urine day and night** and does not void normally. She denies faecal leakage. She has **right foot drop** and has been sent away by her husband. On examination she is pale, with ammoniacal dermatitis. Sims speculum in the left lateral position shows a **2 cm mid-vaginal VVF, with its distal edge 3 cm from the EUM, a clean edge and mild scarring (Goh 2bi)**. The dye test is positive. Rectal exam is normal. **Summary: an ischaemic VVF after prolonged obstructed labour, with foot drop and social isolation.** Plan: Foley catheter on free drainage with fluids of 4–5 L a day, perineal care, a haematinic, physiotherapy, counselling, and referral to the fistula centre for repair once the tissues are clean."

Worked example 1: the three-swab test (Moir)

How to do it:

1. Empty the bladder with a Foley catheter.
2. Place **three swabs (or gauze rolls) in the vagina**, one above the other: **upper** (in the vault, near the cervix), **middle**, and **lower** (just inside the introitus).
3. Instil **about 100–200 mL of dilute methylene blue** into the bladder through the catheter and **clamp it** (or remove it).
4. Ask her to walk around for about **10–15 minutes**.
5. Remove the swabs **in order, from the lowest**, and lay them out.

Finding	Meaning
Upper swab wet but NOT blue (lower swabs dry)	**Ureterovaginal fistula** (clear renal urine bypasses the bladder)

Finding	Meaning
Upper swab blue	High VVF: **juxtacervical / vault or vesicocervical**
Middle swab blue	**Mid-vaginal VVF**
Lowest swab blue only	**Urethrovaginal** fistula, **or** dye leaking from the meatus and tracking back (a false positive): recheck with a speculum
All swabs dry after walking and coughing	No bladder fistula: think **stress, urge or overflow** incontinence (or a ureteric fistula that did not leak during the test; repeat if the story is convincing)
Blue and clear urine both	**Combined** VVF and ureteric fistula

Worked example 2: leaking after a CS, but she voids

A 30-year-old P3 had a repeat emergency CS 2 weeks ago. She now leaks continuously **but also passes normal amounts of urine**. Mild right loin ache; temperature 37.4 °C.

- **Speculum**: urine pooling in the vault; no hole seen.
- **Three-swab test**: upper swab **wet, not blue**; middle and lower swabs dry.
- **Renal ultrasound**: **right hydronephrosis**.

Interpretation: **right ureterovaginal fistula** (surgical ureteric injury at CS). **Plan**: U&E, CT urogram or IVU where available, and **urology referral** for a ureteric stent (retrograde or antegrade) or reimplantation. A Foley catheter alone will not heal it.

> EXAM FAVOURITE
> "Wet but not blue" is the classic viva answer for a **ureterovaginal** fistula, and "continuous leakage with normal voiding" is the classic history.

9. Management

Immediate / emergency

Most women are seen weeks or months later and need no resuscitation. The exceptions are the woman seen days after an obstructed labour or a difficult CS, who may be septic, dehydrated or have urine in her peritoneum. First 5 minutes:

1. **Greet her privately and with dignity** (section 6).
2. **A, B, C, vital signs**: temperature, pulse, BP, RR. Fever, tachycardia or hypotension → look for **puerperal sepsis, pelvic abscess, necrotic slough** or **urinary ascites**; IV access, fluids, cultures, antibiotics, senior surgeon.
3. **Abdomen in 30 seconds**: distension, peritonism (urine or faeces in the peritoneum), a **tender flank** or **loin mass** (obstructed ureter or urinoma), a **palpable bladder** (retention with overflow).
4. **Is she passing any urine at all? Anuria** after pelvic surgery = possible **bilateral ureteric injury** or AKI: urgent U&E, renal ultrasound and a urologist the same day.
5. **Put a Foley catheter in now** (16–18 F, free drainage, never clamped) if the fistula is fresh (within about 4–6 weeks of delivery), or if the bladder is distended. This is both a diagnostic and a therapeutic step.
6. **Screen mood and safety**: ask directly about hopelessness and thoughts of self-harm. Depression is common; suicidal thoughts are a red flag needing same-day support.
7. **Necrotic, foul slough with systemic illness**: debridement and antibiotics.

Definitive

A. The fresh fistula (first 4–6 weeks): conservative management

1. **Insert a Foley catheter (size 16–18 F)** on **continuous free drainage**. Never clamp it. Secure it to the thigh.
2. **High oral fluid intake**: aim for **about 4–5 L a day** so the urine is pale and dilute. This reduces infection, encrustation and blockage.
3. Continue drainage for **up to 4–6 weeks**. Re-examine every 1–2 weeks. **Small fistulae may close**.
4. **Local care**:
   - Remove necrotic slough gently.
   - **Sitz baths** (warm salt water) twice daily.
   - **Barrier cream** (zinc oxide or petroleum jelly) for the vulval skin.
5. **Treat infection** according to culture.
6. **Nutrition**: protein and calories. Give **iron and folate** for anaemia.
7. **Physiotherapy**: for foot drop, and ambulation.
8. **Counselling** and hope: explain that fistula is **curable**, that it is **not a curse**, and that surgery is usually free.
9. If it has not closed by 4–6 weeks, **refer to a fistula centre** for repair once the tissues are healthy.

B. Timing of repair

- **Traditional**: wait **about 3 months** after delivery, so that oedema, infection and necrotic tissue settle and the edges are healthy.
- **Early repair**: many experienced fistula surgeons repair **as soon as the tissues are clean** (no slough, no infection). This is often during the catheter period, and closure rates are excellent. The key rule is **tissue quality, not the calendar**.
- **Iatrogenic injury recognised at operation**: **repair immediately** in layers, and catheterise for about 10–14 days.
- **Iatrogenic injury found after surgery**: a trial of catheterisation. If that fails, repair when the tissues are healthy (commonly after 6–12 weeks). A **ureteric injury** needs a urologist for a stent or reimplantation.

C. Pre-operative preparation

Item	Detail
Nutrition and anaemia	Correct anaemia: ferrous sulphate plus folic acid, and transfuse if Hb is very low
UTI	Treat according to culture
Deworming / schistosomiasis (endemic areas)	Albendazole 400 mg single dose; praziquantel 40 mg/kg single dose if schistosomiasis is suspected
Physiotherapy	For foot drop and contractures
Bowel preparation (RVF or combined)	Low-residue diet or clear fluids, and an enema or bowel preparation per the surgeon
Consent and counselling	Chance of closure, residual incontinence, catheter duration, abstinence, future CS
Pregnancy test	Before surgery
Anaesthesia	**Spinal** for most vaginal repairs
Antibiotic prophylaxis	A single dose at induction (e.g. **ceftriaxone 1 g IV** or **cefazolin 2 g IV**, or gentamicin per the unit's protocol). Routine prolonged antibiotics are not needed

D. Surgical repair principles (VVF)

1. **Route**: the **vaginal route** is used for most obstetric fistulae (well over 90%). The **abdominal route** is used for vesicouterine fistulae, high vault fistulae unreachable from below, ureteric reimplantation, and some complex cases.
2. **Position and exposure**: **exaggerated lithotomy** (or prone or knee–chest for some). Good light and suction. An episiotomy (relaxing incision) if the vagina is narrow.
3. **Identify the ureters** if the fistula is near the trigone. Catheterise the ureters if needed.
4. **Circumscribe** the fistula and **dissect widely** to separate the bladder from the vagina. Mobilise until the bladder edges come together **without tension**.
5. Excise scar sparingly. Do **not** make the hole bigger unnecessarily.
6. **Close the bladder**, usually in **one layer of interrupted absorbable sutures** (e.g. 2-0 or 3-0 polyglactin), transversely or longitudinally, whichever gives less tension.
7. **Test for water-tightness** by instilling dye through the urethral catheter. Re-suture any leak.
8. In scarred or large repairs, consider a **Martius (labial fat-pad) flap** between the bladder and vagina.
9. **Close the vagina** separately, and **do not overlap** its suture line with the bladder line.
10. **Urethral** fistulae may need **urethral reconstruction** and an anti-incontinence procedure (e.g. urethral plication, fascial sling).
11. Insert a **Foley catheter** (plus a suprapubic catheter in some centres) and a **vaginal pack** for 24–48 hours.

E. Post-operative care: "the catheter is the patient"

1. **Uninterrupted bladder drainage**:
   - **Simple fistula**: **7–10 days** (WHO 2018; seven days was as effective as 14).
   - **Complex or urethral repairs**: typically **14 days** or per the surgeon.
2. Nurses check **hourly** that urine is flowing. A **blocked catheter** causes a **ruptured repair**. If flow stops, **gently flush with sterile saline**; don't wait.
3. **Oral fluids of about 4–5 L a day** (or IV if vomiting). Keep a strict **input–output chart**. The urine should be clear.
4. Remove the vaginal pack at 24–48 hours. Keep the perineum clean and dry.
5. **Early ambulation.** Give analgesia. Prevent constipation (lactulose).
6. **After catheter removal**:
   - Timed voiding every 2–3 hours at first.
   - Check residual volume.
   - Start **pelvic floor muscle training**.
   - Observe for leakage (a dye test if in doubt).
7. **Discharge advice**:
   - **No sexual intercourse for about 3 months** (to let the vagina and bladder heal).
   - No heavy lifting.
   - Complete family planning counselling and **offer contraception**.
   - **Any future delivery should be by elective CS** at a hospital.
   - Attend follow-up at 3 months and at 6 months.
8. **Reintegration**: counselling, family mediation, skills training and income support (e.g. TERREWODE and camp programmes).

F. Rectovaginal fistula

1. **Low RVF** (often after a missed or broken-down **third- or fourth-degree tear**):

- a **perineal repair** (convert to a fourth-degree tear and repair in layers, including the **sphincter**), or a transvaginal layered repair
- done when the tissues are clean, usually **about 3 months** after injury

2. **High or large RVF, combined VVF/RVF, or radiation-related RVF**: consider a **diverting colostomy** first, then repair, then close the colostomy.
3. **Around surgery**: bowel preparation, a **low-residue diet**, stool softeners (**lactulose**), and **metronidazole** cover per protocol.
4. **After surgery**: avoid constipation, straining and enemas, and follow up for continence.

G. When closure fails or incontinence persists

Problem	Approach
Failed closure	Re-examine at 3 months. Re-repair at a specialist centre (success falls with each attempt)
Residual stress incontinence (common with urethral involvement)	Pelvic floor muscle training; **urethral plication**, pubococcygeal sling or fascial sling at specialist centres
Irreparable fistula (e.g. total urethral loss, tiny fibrotic bladder)	**Urinary diversion** (e.g. ureterosigmoidostomy / Mainz II pouch) in selected women after careful counselling
Vaginal stenosis	Dilators, flaps
Amenorrhoea / infertility	Evaluate hypothalamic causes and Sheehan's syndrome; offer fertility counselling

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ceftriaxone (surgical prophylaxis)	1 g	IV	Once at induction	Single dose	Or cefazolin 2 g IV, or gentamicin per the unit's protocol; routine prolonged antibiotics not needed
Nitrofurantoin (uncomplicated UTI, per culture)	100 mg	PO	12-hourly	5 days	Avoid if eGFR is low; adjust to the culture result
Metronidazole (RVF / anaerobic cover)	400–500 mg (or 500 mg IV)	PO / IV	8-hourly	Per the surgeon's protocol around RVF repair	Avoid alcohol
Ferrous sulphate + folic acid	200 mg (about 60 mg elemental iron) + 400 µg–5 mg	PO	Once or twice daily	Until anaemia corrected before repair	GI upset, dark stools
Albendazole	400 mg	PO	Once	Single dose	Deworming; avoid in first trimester
Praziquantel (schistosomiasis, endemic areas)	40 mg/kg	PO	Once	Single dose	Treat if haematuria or eggs are found
Lactulose	15 mL	PO	12-hourly	Post-operatively	Soft stools after RVF or perineal repair; avoid straining
Paracetamol	1 g	PO / IV	6-hourly	As needed	Max 4 g/day; add an NSAID if renal function is normal
Zinc oxide cream	Thin layer	Topical	After each wash	Until skin healed	Ammoniacal dermatitis

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Oxybutynin (urgency after repair, if persistent)	2.5–5 mg	PO	2–3 times daily	Per specialist	Anticholinergic side effects; specialist advice

Referral criteria

Referral triggers compiled from the source notes and UCG levels of care Verify *not an MoH referral list*.

Level	What to do	Refer up when
HC II / III	Recognise prolonged labour and **refer early**. Catheterise after any long labour. Identify women leaking urine in the community, reassure them, and **link to a fistula centre or camp**	Any woman with leakage → HC IV / fistula centre; prolonged labour → HC IV for CS
HC IV / general hospital	Timely CS with good surgical technique. **Catheterise for 7–14 days** after obstruction. Manage fresh fistulae with a catheter, local care and nutrition. Trained surgeons may repair **simple** fistulae	Not closed by 4–6 weeks; any **complex** fistula; suspected ureteric injury; sepsis/peritonism after CS → regional/national referral or fistula centre
Regional / national referral hospital, fistula centre	**Complex** repairs, urethral reconstruction, ureteric injuries (urology), RVF with colostomy, diversions, training, and reintegration programmes	Failed repairs, irreparable fistula needing diversion → national specialist centres

Refer to a specialist fistula centre: a complex fistula: large, urethral (Goh 4), circumferential, scarred, previous failed repair, combined VVF/RVF, ureteric or vesicouterine.

Monitoring and follow-up

- **Fresh fistula on catheter**: re-examine every 1–2 weeks; fluids 4–5 L/day; check the catheter drains.
- **After repair**: catheter checked **hourly**; strict input–output chart; remove pack at 24–48 h; catheter 7–10 days (simple) or about 14 days (complex/urethral); after removal, timed voiding, residual volume, dye test if leaking.
- **Follow-up at 3 months and 6 months**: closure, continence, stenosis, mood, contraception, reintegration.
- **Failed closure**: re-examine at 3 months and refer for re-repair.

10. Complications

Disease-related (of the fistula itself)

- Dermatitis.
- Recurrent UTI.
- **Bladder stones**.
- Vaginal stenosis and dyspareunia.
- Amenorrhoea and infertility.
- Foot drop.
- **Depression**, suicide risk, divorce and isolation.
- Malnutrition (women drink less to leak less).

Treatment-related (of surgery)

- Failed closure (about 10–20%).

- **Residual stress incontinence**.
- Haemorrhage.
- **Ureteric ligation**.
- UTI.
- **Catheter blockage** (repair breakdown).
- Vaginal stenosis.
- Urgency or overactive bladder.
- VTE.

In future pregnancy

- Breakdown of the repair if she labours, so she needs **elective CS**.
- A higher risk of obstructed labour recurring (the same pelvis).

11. Prognosis

- Most fistulae can be repaired vaginally, with closure rates of about **80–90%**; in the Ugandan multicentre cohort, closure with continence was about **81%**. Failed closure occurs in about 10–20%.
- Success falls with each repair attempt; **Goh type 4** (urethral involvement), large, circumferential and scarred fistulae have the worst outcomes.
- **Closed is not cured**: many women still have stress incontinence, foot drop, stenosis, infertility and social harm.
- Small fresh fistulae may close with catheter drainage alone.

12. Prevention

Prevention is the most important management.

1. **Primary**:
    - delay the first pregnancy and end child marriage
    - **family planning**
    - **four or more ANC contacts** (the WHO recommends 8) with **birth preparedness**
    - **skilled birth attendance** and facility delivery
    - **maternity waiting homes** for high-risk women
2. **Intrapartum**:
    - **partograph or LCG monitoring**
    - early referral from HC II/III
    - **timely CS** (the decision-to-delivery interval matters)
    - **no oxytocin in obstruction**
3. **At CS**: operate with an **empty bladder** (catheter), make a proper **bladder flap**, identify the ureters in difficult cases, and inspect the bladder before closing (instil dye if in doubt). Repair any recognised injury immediately.
4. **Postpartum**: after **prolonged or obstructed labour**, leave an **indwelling catheter for 7–14 days** with high oral fluids.
5. **After repair**: elective CS for every future delivery; contraception until healed.
6. **System**: document the three delays and review each case (MPDSR-style).

13. Special Populations

- **Adolescents and child brides**: small pelvis, highest risk of obstructed labour; counselling, family planning and reintegration are central.
- **Pregnancy after repair**: elective CS in hospital; labour risks breakdown of the repair and recurrent obstruction.
- **Women with HIV**: check status, start or continue ART; HIV and malnutrition affect healing.
- **Schistosomiasis-endemic areas**: urine for ova; praziquantel 40 mg/kg before surgery.
- **Radiation or cancer-related fistula**: biopsy; oncology and palliative care, often diversion or colostomy rather than standard repair.

14. Key Points

High-yield points

- **Ischaemic fistula leaks 3–10 days** after delivery; **iatrogenic** leaks immediately; **ureteric** 1–3 weeks after CS/hysterectomy.
- **Continuous leakage + normal voiding = ureterovaginal**; **cyclical haematuria + amenorrhoea + continence = vesicouterine** (Youssef).
- **Dye test before calling it a fistula**: rule out overflow (residual >150 mL) and stress incontinence.
- **Three-swab test**: upper wet-not-blue = ureteric; middle blue = mid-vaginal; upper blue = vault; lowest only = urethral or spill.
- **Catheter after every long labour (7–14 days)**; fresh fistula: Foley 16–18 F, free drainage, up to 4–6 weeks, 4–5 L fluids.
- **Goh**: site (EUM distance), size, scarring; type 4 is worst.
- Repair: exposure, mobilisation, tension-free watertight separate layers, Martius flap if scarred, **uninterrupted drainage**.
- WHO 2018: catheter **7–10 days** after simple repair.
- After repair: no intercourse 3 months, contraception, **elective CS** for every future baby.
- Never skip the **rectal examination**; biopsy a hard friable "fistula".

Numbers & doses

Item	Value
Slough separates (ischaemic fistula leaks)	**3–10 days** after delivery
Catheter after prolonged or obstructed labour (prevention)	**7–14 days**, high fluids
Fresh fistula: continuous catheter drainage	**Up to 4–6 weeks**, Foley **16–18 F**, never clamped
Oral fluids with a catheter (fresh fistula and after repair)	About **4–5 L/day**
Dye test volume	About **100–200 mL** dilute methylene blue
Three-swab test walk	**10–15 minutes**
Goh site (distal edge to EUM)	Type 1 **>3.5 cm**; 2 **2.5–3.5 cm**; 3 **1.5–<2.5 cm**; 4 **<1.5 cm**
Goh size	a **<1.5 cm**; b **1.5–3 cm**; c **>3 cm**
Goh scarring i	Vaginal length **over 6 cm**, normal capacity
Simple fistula	Single, under about **3–4 cm**, no urethral closing mechanism, first repair

Item	Value
Catheter after simple repair	**7–10 days** (WHO 2018); complex or urethral about **14 days**
Catheter after iatrogenic injury repaired at operation	About **10–14 days**
Traditional timing of delayed repair	About **3 months** (but tissue quality, not the calendar, decides)
Closure rates	About **80–90%**; Ugandan cohort closure with continence about **81%**; failure **10–20%**
No intercourse after repair	About **3 months**
Post-void residual suggesting retention	**Over 150 mL**
Surgical prophylaxis	**Ceftriaxone 1 g IV** (or cefazolin 2 g IV) single dose at induction
Praziquantel (schistosomiasis)	**40 mg/kg** orally, single dose
Albendazole	**400 mg** single dose
New fistulae worldwide	**50,000–100,000** a year
Uganda	About **100,000–200,000** living with fistula; about **1,900** new cases a year; **66%** ischaemic
Future deliveries	**Elective CS** in hospital

Exam traps

- **"Leaking since delivery = VVF."** Not until a **dye test** is positive; overflow from retention and stress incontinence look similar.
- **A negative dye test excludes a fistula.** No: continuous leakage with a negative (clear) test is a **ureterovaginal** fistula until proven otherwise.
- **"Send her home and come back in 3 months."** Wrong for a fresh fistula: **catheterise now**; a small fresh fistula may close.
- **Clamping the catheter** or "bladder training" after repair. The catheter must drain **continuously**; a blocked catheter blows the repair.
- **Forgetting the rectal examination** and missing an RVF or sphincter injury.
- **Labelling a hard, friable, bleeding "fistula" as obstetric.** Biopsy: cervical cancer or radiation.
- **Closed = cured.** Many women still have **stress incontinence** (urethral involvement), foot drop, stenosis, infertility and social harm.
- **Future vaginal delivery after repair.** No: **elective CS**.

Questions seniors ask

Viva questions

1. **Why does leakage start days after delivery in an ischaemic fistula?** The necrotic tissue takes 3–10 days to slough.
2. **How do you tell a ureterovaginal fistula from a VVF?** Normal voiding plus continuous leakage. On the three-swab test the swabs are wet but not blue.
3. **What is Youssef syndrome?** A vesicouterine fistula: cyclical haematuria (menouria), amenorrhoea, and continence.
4. **What are the principles of repair?** Exposure, wide mobilisation, tension-free watertight layered closure, non-overlapping suture lines, well-vascularised tissue (Martius), and **uninterrupted catheter drainage**.
5. **How long is the catheter after simple fistula repair?** 7–10 days (WHO 2018; 7 days was as good as 14).
6. **What advice about future pregnancy?** Elective CS at a hospital.

7. **What is the obstructed labour injury complex?** The ischaemic field injury: VVF/RVF, urethral loss, vaginal stenosis, foot drop, amenorrhoea, infertility, and social consequences.

On the ward round

- **How do you do and read a three-swab test?** Three vaginal swabs, methylene blue in the bladder, walk for 10–15 minutes, remove from below. Upper wet-not-blue = ureteric; middle blue = mid-vaginal VVF; upper blue = juxtacervical/vault; lowest only = urethral (or spill).
- **What do you measure on speculum examination?** Distance of the distal edge from the EUM, size, scarring and vaginal length, urethral length, edges (slough vs clean), plus a rectal exam: this gives the **Goh** grade.
- **Why does timing of onset matter?** Immediate = surgical injury; 3–10 days = ischaemic slough; 1–3 weeks after CS/hysterectomy = ureter.
- **What would make you refer to a specialist fistula centre?** A complex fistula: large, urethral (Goh 4), circumferential, scarred, previous failed repair, combined VVF/RVF, ureteric or vesicouterine.
- **What is Youssef syndrome?** Vesicouterine fistula after CS: cyclical haematuria (menouria), amenorrhoea, continence.
- **What is the most important nursing task after repair?** Keeping the catheter draining **every hour**, with 4–5 L fluids and a strict input–output chart.
- **What is the cheapest fistula prevention?** Timely CS, and a catheter for 7–14 days after every prolonged or obstructed labour.

Mnemonics

- **Timing tells the cause: "Immediate = Iatrogenic, a week = Ischaemic, two weeks after hysterectomy = Ureter."**
- **Repair: "TEAR-proof". T**ension-free, **E**xposure, **A**dequate mobilisation, **R**eal watertight test, **P**roper (separate, non-overlapping) layers, **R**ich blood supply (Martius), **O**utflow uninterrupted (catheter), **O**ral fluids 4–5 L, **F**ollow-up and family planning.
- **Goh: "Site–Size–Scar"**: numbers 1–4 for site (4 = urethra, worst), letters a–c for size, Roman numerals i–iii for scarring.
- **"Catheter after every long labour"**: the cheapest fistula prevention in the world.

OSCE tips

- In a counselling station, **start with empathy and dignity**. Use the words "it is not your fault" and "it can be treated". Explain the catheter, the surgery, abstinence, and future CS. **Ask about mood and support.**
- In a history station, always ask about **faecal leakage, voiding, foot weakness, periods, and social consequences**.
- In an examination or dye-test station: Sims speculum in left lateral, measure the distance from the EUM, size and scarring, pass a sound up the urethra, do a rectal examination, then perform and interpret the dye or three-swab test.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define obstetric fistula.	An abnormal communication between the genital tract and the urinary tract (**VVF**, urethrovaginal, vesicouterine, ureterovaginal) or the rectum (**RVF**), arising from childbirth. It is usually ischaemic after prolonged obstructed labour, or iatrogenic at CS or hysterectomy.
2	Mechanism of an ischaemic obstetric fistula?	The impacted fetal head compresses the bladder base, urethra, vagina (and rectum) against the pubis or sacrum, causing **ischaemic pressure necrosis**. The slough separates at **3–10 days**, leaving a hole.
3	What does the timing of leakage onset tell you?	**Immediately after surgery**: direct surgical injury. **3–10 days postpartum**: ischaemic (obstructed labour). **1–3 weeks after hysterectomy or CS**: ureteric injury.
4	Continuous leakage but normal voiding: think?	**Ureterovaginal fistula**. One ureter drains into the vagina; the other still fills the bladder.
5	What is Youssef syndrome?	A **vesicouterine fistula** (usually after CS): **cyclical haematuria (menouria)**, amenorrhoea, and usually **urinary continence**.
6	How is the dye test done, and how is it interpreted?	Instil dilute **methylene blue** (about 100–200 mL) into the bladder via a catheter, with swabs in the vagina. **Blue = VVF. Wet but not blue = ureterovaginal** (three-swab test).
7	Double-dye test?	Oral **phenazopyridine** turns renal urine **orange**. **Methylene blue** in the bladder turns bladder urine **blue**. Orange on the swab = ureteric; blue = bladder; both = combined.
8	Goh classification: the site types.	Distance of the distal edge from the **EUM**: type 1 more than **3.5 cm**, type 2 = 2.5–3.5 cm, type 3 = 1.5 to under 2.5 cm, type 4 under **1.5 cm** (urethral, worst prognosis).
9	Goh size and scarring grades?	**Size**: a under 1.5 cm, b 1.5–3 cm, c over 3 cm. **Scarring**: i none or mild (vagina over 6 cm), ii moderate or severe fibrosis or a short vagina, iii special (radiation, ureteric, circumferential, previous repair).
10	Waaldijk classification?	**I**: urethral closing mechanism not involved. **II**: involved (A without or B with (sub)total urethral involvement; a without or b with a circumferential defect). **III**: ureteric or other exceptional fistulae.
11	Immediate management of a fresh VVF?	A **Foley catheter (16–18 F) on continuous free drainage** for **up to 4–6 weeks**, oral fluids **4–5 L a day**, sitz baths and barrier cream, nutrition, iron, treat UTI, physiotherapy, counselling. Small fresh fistulae may close.
12	What is the key principle on the timing of repair?	**Tissue quality, not the calendar.** Traditionally wait about 3 months. Many surgeons repair as soon as the tissues are clean, with no slough or infection. Repair a recognised intra-operative injury immediately.
13	Principles of VVF repair?	Good **exposure**, **wide mobilisation**, **tension-free**, **watertight** closure (dye test), **separate, non-overlapping layers**, a well-vascularised flap (Martius) if scarred, and **uninterrupted bladder drainage** afterwards.
14	How long should the catheter stay after a simple VVF repair (WHO 2018)?	**7–10 days**. Short-term (7-day) catheterisation was as effective as longer drainage for a simple fistula. Complex or urethral repairs usually need about 14 days.
15	What is the single most dangerous post-operative event after VVF repair?	A **blocked catheter**: bladder distension disrupts the repair. Check the flow hourly, keep oral fluids high, and flush gently if it stops.

#	Question	Answer
16	Discharge advice after fistula repair?	No intercourse for about **3 months**, no heavy lifting, pelvic floor exercises, family planning, follow-up at 3 and 6 months, and **elective CS** for any future delivery.
17	Components of the obstructed labour injury complex?	VVF/RVF, urethral loss, vaginal stenosis, **foot drop**, pubic osteitis, amenorrhoea or infertility, dermatitis, bladder stones, depression, divorce and poverty.
18	Most important preventive measure after prolonged obstructed labour?	An **indwelling catheter for 7–14 days** with high oral fluids, plus timely CS so it never gets that far.
19	Management of a low RVF after a broken-down fourth-degree tear?	Repair when the tissues are clean (about 3 months). Convert it to a fourth-degree tear and repair it in layers, including the **anal sphincter**. Bowel preparation, stool softeners and a low-residue diet. A colostomy is for large, high or complex fistulae.
20	Name Ugandan fistula services.	About **29 fistula centres in 24 districts**. Regional and national referral hospitals, **Kitovu Hospital** camps, **TERREWODE** Women's Community Hospital (Soroti), and the **Fistula Foundation Treatment Network** with the MoH. Repair is usually free.
21	How do you perform the three-swab (Moir) test?	Empty the bladder; place **three swabs** in the vagina (upper, middle, lower); instil **about 100–200 mL dilute methylene blue** into the bladder and clamp; she walks for **10–15 min**; remove the swabs **from below upwards** and inspect each.
22	Three-swab test: interpret each result.	**Upper wet, not blue** = ureterovaginal. **Upper blue** = juxtacervical/vault (vesicocervical). **Middle blue** = mid-vaginal VVF. **Lowest only blue** = urethrovaginal or dye spilling from the meatus. **All dry** = no bladder fistula: think stress, urge or overflow incontinence.
23	A woman leaks after delivery. Which bedside checks exclude a fistula mimic?	Palpate for a **distended bladder** and measure the **post-void residual** (over 150 mL = retention with overflow); a **cough stress test** (stress incontinence); **dipstick** (UTI); and a **dye test** (negative in all three mimics).
24	Red flags in the first weeks after CS or hysterectomy in a woman with leakage?	**Fever, peritonism, loin pain or a flank mass, anuria**: bladder or bowel injury with leakage into the peritoneum, or ureteric obstruction/urinoma. Urgent U&E, **renal ultrasound**, senior surgeon and **urology**.
25	Which imaging for a suspected ureterovaginal fistula, and what does it show?	**Renal ultrasound** (hydronephrosis, urinoma), then **IVU or CT urogram** (contrast leak, obstruction) at a regional/national referral hospital. Management is **urological**: stent or reimplantation; a Foley alone will not heal it.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 19-year-old para 1 delivered a stillborn baby after 3 days of labour at home. On day 7 she noticed continuous leakage of urine. She presents on day 12. The speculum shows a 1 cm mid-vaginal VVF with sloughy edges. What is the MOST appropriate initial management?

A Immediate vaginal repair today
B Continuous Foley catheter drainage with high oral fluid intake, local care and review
C Send her home and book a repair in 6 months
D Urinary diversion (ureterosigmoidostomy)
E Oral antibiotics and pelvic floor exercises only

2. A 38-year-old had a total abdominal hysterectomy for fibroids. Two weeks later she reports continuous watery vaginal leakage but still passes urine normally. The methylene blue bladder test shows wet vaginal swabs that are not blue. What is the MOST likely diagnosis?

A Vesicovaginal fistula
B Stress urinary incontinence
C Vaginal cuff infection
D Ureterovaginal fistula
E Vesicouterine fistula

3. A 30-year-old who had two previous CS reports monthly blood in her urine since her last CS 8 months ago. She has had no vaginal periods and is continent of urine. What is the MOST likely diagnosis?

A Vesicouterine fistula (Youssef syndrome)
B Bladder endometriosis
C Asherman syndrome
D Schistosomiasis haematobium
E Bladder carcinoma

4. Three days after a simple VVF repair, the nurse notes no urine in the catheter bag for 2 hours. The patient has lower abdominal discomfort. What is the MOST appropriate immediate action?

A Give IV furosemide
B Remove the catheter and ask her to void
C Check the tubing for kinks and gently flush the catheter with sterile saline to restore drainage
D Reassure her and review in the morning
E Clamp the catheter to train the bladder

5. According to the WHO 2018 recommendation, what is the appropriate duration of bladder catheterisation after surgical repair of a simple obstetric urinary fistula?

A 24–48 hours
B 3 days
C 21 days
D 42 days
E 7–10 days

6. A fistula surgeon records: 'distal edge 1 cm from the external urethral meatus, 3.5 cm diameter, dense fibrosis, previous failed repair.' Which statement is MOST accurate?

A This is Goh type 1ai, with an excellent prognosis
B This is a Goh type 4 fistula with a poor prognosis for continence, and needs a specialist centre
C This can be repaired by any medical officer at an HC IV
D This fistula will close with catheter drainage alone
E Classification has no prognostic value

7. During an emergency CS for obstructed labour at an HC IV, the medical officer notices a 2 cm tear in the bladder dome after delivering the baby. What is the BEST management?

A Close the abdomen and refer for fistula repair in 3 months
B Pack the pelvis and transfer immediately
C Repair the bladder in two layers with absorbable sutures, check it is watertight, and leave a catheter for about 10–14 days
D Insert a suprapubic drain only
E Leave it to heal and remove the catheter after 24 hours

8. Which history finding BEST suggests that a woman's postpartum incontinence is overflow rather than a fistula?

A Dribbling with a palpably distended bladder and a large residual on catheterisation
B Leakage onset on day 7
C Continuous leakage day and night
D Faeces per vaginam
E Blue dye on vaginal swabs

9. A 20-year-old woman had a successful VVF repair 6 months ago. She is now 12 weeks pregnant. What is the MOST appropriate delivery plan?

A Vaginal delivery with an early episiotomy
B Home delivery with a TBA
C Induction of labour at 37 weeks
D Elective caesarean section at a hospital near term
E Vacuum delivery to shorten the second stage

10. Which is the MOST effective health-system intervention to prevent obstetric fistula in Uganda?

A Routine episiotomy for all primigravidas
B Prophylactic antibiotics after every vaginal delivery
C Pelvic floor exercises in pregnancy
D Routine vaginal packing after delivery
E Skilled attendance with labour monitoring (partograph or LCG) and timely access to caesarean section

11. A woman has constant urine leakage since a prolonged labour 4 months ago. The dye test confirms a 2 cm VVF 4 cm from the EUM with clean edges. Her Hb is 7.8 g/dL and urinalysis shows nitrites. What should you do BEFORE surgery?

A Operate immediately to avoid further delay
B Correct the anaemia and treat the UTI according to culture, then schedule the repair
C Give prophylactic antibiotics for 6 weeks
D Wait another 12 months
E Perform urinary diversion

12. A woman whose VVF was repaired 3 months ago has a closed fistula (negative dye test) but leaks when she coughs or stands up. The original fistula involved the bladder neck. What is the MOST likely explanation?

A Recurrent VVF
B Ureterovaginal fistula
C Residual stress urinary incontinence from damage to the urethral closing mechanism
D Overflow incontinence
E Detrusor overactivity from UTI

13. A 25-year-old reports passage of flatus and faecal material per vaginam 2 months after a vaginal delivery complicated by a perineal tear that was 'stitched in the ward'. The speculum shows a 0.5 cm opening just above the posterior fourchette. The anal sphincter is deficient anteriorly. What is the MOST appropriate plan?

A Immediate repair today in the labour ward
B Diverting colostomy
C Metronidazole only; it will heal
D Vaginal packing
E Delayed layered repair (including the sphincter) when the tissues are healthy, with bowel preparation and stool softeners

14. A woman with a repaired VVF asks when she may resume sexual intercourse. What is the BEST advice?

A After about 3 months, once healing is confirmed at follow-up
B Immediately after catheter removal
C After 1 week
D Never again
E Only after her next delivery

15. Which pre-operative finding would MOST strongly lead you to choose an abdominal rather than a vaginal approach for a urinary fistula repair?

A A 1 cm mid-vaginal VVF
B A juxta-urethral fistula
C Mild vaginal scarring
D A vesicouterine fistula with cyclical haematuria after CS
E A fistula 4 cm from the EUM

Answers & explanations

> **1. B** · Continuous Foley catheter drainage with high oral fluid intake, local care and review
>
> A **small fresh fistula** should be managed with **continuous catheter drainage** (up to 4–6 weeks), fluids of 4–5 L a day, and perineal care. It may close spontaneously, and the tissues improve for repair. **A** Sloughy, necrotic edges do not hold sutures. **C** Sending her away without a catheter wastes the chance of spontaneous closure. **D** Diversion is only for irreparable fistulae. **E** Pelvic floor exercises do not close a fistula.

> **2. D** · Ureterovaginal fistula
>
> Continuous leakage with **normal voiding**, and **wet but not blue** swabs after hysterectomy, indicate a **ureterovaginal fistula**. **A** A VVF would stain the swabs blue. **B** SUI is effort-related, not continuous. **C** Cuff infection gives a purulent discharge, not urine. **E** There is no uterus after hysterectomy.

3. A · Vesicouterine fistula (Youssef syndrome)

Cyclical haematuria, amenorrhoea and continence after CS is **Youssef syndrome**. **B** Endometriosis can cause cyclical haematuria, but her vaginal periods would continue. **C** Asherman gives amenorrhoea without haematuria. **D** Schistosomiasis haematuria is not cyclical. **E** Carcinoma is unlikely in a young woman with this cyclical pattern.

4. C · Check the tubing for kinks and gently flush the catheter with sterile saline to restore drainage

A **blocked catheter** risks bladder distension and **breakdown of the repair**. Restore drainage immediately: check for kinks and gently flush (or replace per the surgeon's protocol). **A** Furosemide makes more urine against a blockage. **B** Early removal endangers the repair. **D** Delay is dangerous. **E** Clamping is contraindicated.

5. E · 7–10 days

WHO (2018) recommends **short-term catheterisation (7–10 days)** for simple fistula repair. A trial showed 7 days was as effective as 14, with less infection and cost. **A/B** are too short for bladder healing. **C/D** are unnecessarily long and add infection and cost.

6. B · This is a Goh type 4 fistula with a poor prognosis for continence, and needs a specialist centre

Under 1.5 cm from the EUM is **type 4**; over 3 cm is **c**; previous repair or dense fibrosis is **ii/iii**. Urethral involvement predicts failure and **residual stress incontinence**, so this is a **complex** fistula for a specialist. **A** Wrong grading. **C** Complex fistulae need expert surgeons. **D** A large, scarred fistula won't close with a catheter. **E** Goh's system predicts closure.

7. C · Repair the bladder in two layers with absorbable sutures, check it is watertight, and leave a catheter for about 10–14 days

A recognised intra-operative bladder injury should be **repaired immediately** in layers, tested for water-tightness, and drained with a catheter (typically 10–14 days). **A/E** Leaving it guarantees a fistula or urinary peritonitis. **B** Transfer is unnecessary for a simple dome tear and adds risk. **D** A drain alone does not close the defect.

8. A · Dribbling with a palpably distended bladder and a large residual on catheterisation

Overflow incontinence (postpartum retention) gives dribbling with a **distended bladder** and a large residual. It is treated with catheter drainage. **B/C** suggest an ischaemic fistula. **D** suggests an RVF. **E** confirms a VVF.

9. D · Elective caesarean section at a hospital near term

After fistula repair, **elective CS** protects the repair from breakdown and avoids a repeat obstructed labour. **A/C/E** involve labour, which risks recurrence. **B** Home delivery is dangerous for anyone with this history.

10. E · Skilled attendance with labour monitoring (partograph or LCG) and timely access to caesarean section

Fistula results from **unrelieved obstructed labour**. Recognising prolonged labour and **timely CS** prevents it. **A** Episiotomy does not relieve obstruction. **B** Antibiotics do not prevent pressure necrosis. **C** Pelvic floor exercises help prolapse and SUI, not fistula. **D** Packing is irrelevant.

11. B · Correct the anaemia and treat the UTI according to culture, then schedule the repair

Optimise her first: treat the UTI and correct the anaemia (haematinics, or transfusion if needed), along with nutrition and counselling. Then repair. The fistula (Goh 1bi) is favourable. **A** Anaemia and infection impair healing. **C** Prolonged prophylaxis is unnecessary. **D** There is no reason to wait a year. **E** Diversion is for irreparable fistulae.

12. C · Residual stress urinary incontinence from damage to the urethral closing mechanism

When the fistula involved the **urethra or bladder neck**, the continence mechanism is damaged, causing **residual SUI** despite closure. It is treated with pelvic floor exercises and a plication or sling procedure. **A** The dye test is negative. **B** Leakage would be continuous. **D** Overflow gives dribbling with a full bladder. **E** Urgency, not effort, would be the trigger.

13. E · Delayed layered repair (including the sphincter) when the tissues are healthy, with bowel preparation and stool softeners

This is a **low RVF after an unrecognised or broken-down fourth-degree tear**. The treatment is **delayed layered repair** (convert to a fourth-degree tear, repair the rectal mucosa, **sphincter** and perineum) in theatre with good tissues, bowel preparation and softeners. **A** Repair in a ward with inflamed tissue fails. **B** A colostomy is for high, large or complex RVFs. **C/D** won't close an epithelialised fistula.

14. A · After about 3 months, once healing is confirmed at follow-up

Abstain for **about 3 months** to allow full healing of the bladder and vaginal repair. **B/C** are too early and risk breakdown. **D** is unnecessary and harmful to her marriage and wellbeing. **E** makes no sense, and a future delivery should be by CS.

15. D · A vesicouterine fistula with cyclical haematuria after CS

Vesicouterine fistulae (and high vault or ureteric fistulae) are usually repaired **abdominally**. **A/B/C/E** are all typically approached **vaginally**, the route for most obstetric fistulae.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The fresh fistula on the postnatal ward

A 17-year-old para 1+0 is on the postnatal ward of a regional referral hospital, day 8 after a CS for obstructed labour following 2 days of labour at home. The baby was stillborn. The catheter was removed on day 2. Today she is wet all the time and says urine 'just flows'.

1. What history do you take?

- **Onset** of leakage (day 6–8 suggests an ischaemic slough).
- Continuous vs effort-related.
- Whether she **voids** at all.
- **Faecal or flatus leakage**.
- Fever or loin pain.
- **Leg weakness** (foot drop).
- Mood and support: husband, family.
- Future pregnancy wishes.
- HIV status and nutrition.

2. How do you examine and confirm the diagnosis?

- Look for **ammoniacal dermatitis**.
- Palpate the abdomen for a distended bladder (to exclude overflow).
- **Sims speculum** in the left lateral position, with a good light: locate the fistula, size it, measure its distance from the EUM, and note slough and scarring.
- **Dye test**: methylene blue via catheter with a vaginal swab.
- **Rectal exam** for RVF.
- Check the foot for dorsiflexion.

3. You find a 1.5 cm sloughy VVF 3 cm from the EUM. What is your management now?

1. **Re-insert a Foley 16–18 F on free drainage** (never clamp).
2. Oral fluids **4–5 L a day**; chart input and output.
3. Sitz baths and zinc oxide cream.
4. Gentle removal of slough.
5. Urinalysis and culture, and treat UTI.
6. Hb, then haematinics.
7. Nutrition.
8. Physiotherapy.
9. **Counselling**: this is common, treatable, and not her fault.
10. Re-examine weekly for **up to 4–6 weeks**.

4. At 5 weeks the fistula is 1 cm with clean edges and has not closed. What next?

Refer to the fistula surgeon or regional centre (or the next camp) for repair now that the tissues are clean. Classify it (e.g. Goh 2ai). Optimise her Hb, treat any UTI, and give single-dose antibiotic prophylaxis at spinal anaesthesia. Plan a vaginal layered tension-free repair with a dye check.

5. What post-operative and discharge instructions will you give?

> - Catheter for **7–10 days** (simple fistula), checked hourly for flow.
> - Oral fluids of 4–5 L a day.
> - Early ambulation.
> - After removal: timed voiding and pelvic floor exercises.
> - **No intercourse for about 3 months**; no heavy lifting.
> - **Family planning**.
> - **Elective CS** in any future pregnancy.
> - Follow-up at 3 and 6 months.
> - Reintegration support.

Take-away: A fresh fistula is an opportunity: catheterise continuously with high fluids and good local care. Many small ones close or become ideal for repair. The catheter is the patient.

Case 2: Leaking after hysterectomy

A 44-year-old had an emergency peripartum hysterectomy for uterine rupture at a general hospital. Ten days after discharge she returns with continuous clear vaginal leakage, but she says she still 'passes urine like before'. She also has right loin ache.

1. What is your leading differential, and why?

> **Right ureterovaginal fistula**: a ureteric injury at hysterectomy (the ureter lies close to the uterine artery and cardinal ligament). **Normal voiding** means the left ureter still fills the bladder. The differential includes a **VVF** (vault) or a combined fistula. Loin ache suggests partial obstruction or hydronephrosis.

2. Which bedside test helps, and what are the possible results?

> The **three-swab (tampon) test** with methylene blue in the bladder:
>
> - **Wet but not blue**: ureteric.
> - **Blue**: VVF.
>
> A **double-dye test** (oral phenazopyridine plus bladder methylene blue) shows orange (ureteric), blue (bladder), or both.

3. What investigations follow?

> - **Renal ultrasound** (hydronephrosis, urinoma).
> - Urea, creatinine and electrolytes.
> - Urinalysis and culture.
> - **IVU or CT urography** to localise the leak and assess the other kidney.
> - Cystoscopy with retrograde pyelography at a urology or referral centre.

4. How is it managed?

> **Refer to urology / the fistula centre.**
>
> - Attempt a **retrograde or antegrade ureteric stent (JJ)**. Small injuries may heal over 6 weeks.
> - If obstructed and a stent fails: **percutaneous nephrostomy**, then a **ureteric reimplantation** (ureteroneocystostomy, with a psoas hitch or Boari flap if needed), usually by the **abdominal route**.
> - Treat infection and protect renal function.

5. How could this have been prevented?

- Identify the ureters during difficult hysterectomy (palpate or visualise, especially near the uterine vessels and cardinal ligament).
- Clamp close to the uterus.
- Inspect for injury before closing.
- Senior help early.
- Where suspected, give IV methylene blue or indigo carmine, do cystoscopy, and repair immediately if found.

Take-away: Continuous leakage with normal voiding after pelvic surgery means ureterovaginal fistula until proven otherwise. The dye test is negative, the swabs are wet. It is a urology problem.

Case 3: Counselling the rejected woman

A 26-year-old woman from Teso is brought by a community health worker to a fistula outreach camp. She has leaked urine and stool for 4 years since a 3-day labour that ended in a stillbirth. Her husband left her. She walks with a high-stepping gait and has not menstruated since.

1. What components of the obstructed labour injury complex should you look for?

- VVF and **RVF** (combined).
- Urethral involvement.
- Vaginal stenosis.
- **Foot drop** (high-stepping gait).
- Amenorrhoea: hypothalamic, or Sheehan's if there was PPH.
- Dermatitis.
- Bladder stones.
- Malnutrition.
- **Depression**, suicidal thoughts, social exclusion.

2. Examination shows a 4 cm juxta-urethral VVF with dense fibrosis and a 2 cm mid-rectovaginal fistula. How do you classify it and where should she be treated?

A **complex combined fistula**. It is **Goh type 4 (or 3), c, ii** (under about 2.5 cm from the EUM, over 3 cm, dense fibrosis), plus an RVF. She needs a **specialist fistula centre** (e.g. a regional or national referral hospital, or TERREWODE / a partner centre), not a general HC IV.

3. What pre-operative preparation is needed?

- Nutrition and correction of anaemia.
- Treat UTI.
- Deworming, and praziquantel if schistosomiasis is endemic or haematuria is present.
- **Physiotherapy** for foot drop.
- **Bowel preparation**, and possibly a **diverting colostomy** for the RVF first.
- **Psychosocial counselling**.
- Consent covering realistic outcomes: closure is likely, but continence is less certain; there may be more than one operation; urinary diversion is a last resort.

4. How do you counsel her about her future?

- **Empathy**: it is not her fault or a curse.
- Treatment is free.
- Residual stress incontinence is possible, and further procedures (sling, plication) exist.
- Abstain from intercourse for about 3 months after repair.
- If she conceives, she needs **elective CS** at a hospital.
- Contraception.
- **Reintegration** programmes: skills and livelihood, peer support, and family mediation where she wishes.

Take-away: Fistula is a social catastrophe as much as a surgical one. Treat the whole woman: the fistulae, the foot drop, nutrition, mood and reintegration, and route complex cases to specialist centres.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.